

Photon sector (null worldlines)

no proper time: $d\tau = 0$, no records written

$\beta_\gamma < 0.0039$ (95 %): acoustic-scale fit

$\beta_\gamma/\beta_m < 0.025$

$H_0 = 67.16 \pm 0.47$ (Level 2 chain)

$\Sigma = 1$: lensing and θ_s unmodified

Matter sector (timelike worldlines)

records written: $S_{\text{info}} > 0$ iff $d\tau > 0$

$\beta_m = \Omega_m/2 = 0.15765$, fixed

$\mu(a) < 1$: $\mu_0 = -0.136$, σ_8 0.809 $\rightarrow$ 0.800

$H_0^{\text{matter}} = 67.16\sqrt{1 + \beta_m} = 72.26$

background (distances): LCDM

Type Ia supernovae (Pantheon+)

Hubble-flow shape: $\beta = -0.035$ (68 %: -0.065 to 0)

β_m in the distances: $\Delta\chi^2 = +23.6$

M free: magnitudes carry no H_0

normalisation: Cepheid ladder 73.04 ± 1.04

Mechanism (Chapter on the theory)

irreversible records on timelike worldlines

Landauer cost on the horizon: $-dE = T_H d(S_{\text{geo}} + S_{\text{info}})$

$E(a) = \exp(1 - 1/a)$, production per e-fold peaks today

no free parameter beyond Λ CDM